Vishal S

github.com/vishals25 | linkedin.com/in/vishals25 | vishalsubramaniam0@gmail.com | +91-99522 53903

Summary

Computer Science graduate and software engineer with Python, SQL databases and LLM tooling (LangChain, Pinecone, Hugging Face, Claude API), applied machine learning for geospatial analysis, and production Ruby on Rails and Python API (Django, FastAPI) experience.

Education

Coimbatore Institute of Technology May 2026
Bachelor of Engineering in Computer Science **CGPA: 8.86**

Skills

Programming Languages: Ruby, Python, JavaScript, TypeScript, C

AI & ML: LangChain, Claude API, Pinecone (vector database), Hugging Face, Scikit-learn, Autoencoders

Frontend: HTML, CSS, Tailwind CSS, React, Streamlit

Web Frameworks: Ruby on Rails, Django, Flask, FastAPI, React, AngularJS, NodeJS

Databases: PostgreSQL, MySQL, SQLite3, Firebase Firestore

DevOps & Cloud: Docker, Kubernetes, Argo CD, GitHub Actions, Trivy

Tools: Git, GitHub, GitLab, Azure DevOps, Jira

Experience

Rootquotient Technologies **Dec 2025 – Present**
SDE (Jun 2026 – Present) | SDE Intern (Dec 2025 – May 2026) *Chennai, India*

- Engineered and delivered production features on a legacy Ruby on Rails enterprise pension application, including cron job scheduling, MFA integration and Ping Authentication workflows serving thousands of plan members.
- Redesigned the pension application flow with targeted UX improvements, incresing submissions by over 75%.
- Supported Docker-based containerization of the on-premise environment, coordinating across database, infrastructure, and DevOps dependencies via VPN and remote desktop to ensure zero-downtime delivery.

Sálim Ali Centre for Ornithology and Natural History (SACON) **Jan 2025 – Feb 2025**
Software Engineering Intern *Coimbatore, India*

- Developed a machine learning approach to habitat suitability analysis for hornbills using autoencoders and geospatial visualization.
- Processed satellite and field data to create interactive maps identifying priority conservation areas.
- Integrated computational methods with ecological research to support biodiversity conservation.

Projects

Tetris GitOps Pipeline | *Docker, GitHub Actions, Trivy, Kubernetes, Argo CD, Kustomize | GitHub* **Aug 2026**

- Built a CI/CD and GitOps pipeline around a forked open-source JavaScript game: GitHub Actions auto-versions releases, scans with Trivy, and publishes the image to Docker Hub.
- Deployed to a Kubernetes cluster with Argo CD and Kustomize using a pull-based model, so no cluster credentials are stored in GitHub; rollbacks are done through Git.
- Hardened the container (non-root user, read-only filesystem, dropped capabilities) and documented the architecture and runbook.

Avian-Airplane Collision Prediction | *Python, YOLOv8, LSTM, TensorFlow Lite* **Nov 2025**

- Designed and Developed a deep learning framework that detects birds in airport video with a fine-tuned YOLOv8 model and forecasts their flight paths with an LSTM.
- Tested the prediction model to confirm it runs fast enough for live video, and mapped bird activity around Coimbatore airport from eBird sightings to show where the highest-risk areas are.

Hornbill Habitat Suitability Analysis | *Python, Scikit-learn, Autoencoders, GeoPandas | GitHub* **Mar 2025**

- Built a machine learning pipeline to model habitat suitability of hornbill species using ecological, telemetry, and climate data.
- Identified core habitat zones and movement behavior, and visualized spatial patterns on interactive maps with satellite imagery layers.

Fun-Gram | *Vite, React, Chakra UI, Firebase | GitHub* **Jan 2025**

- Built an Instagram-style social media app with real-time posting, liking, and commenting.
- Integrated Firebase Authentication, Firestore, and Storage; designed a responsive UI with Chakra UI and deployed on Firebase Hosting.